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The Sabaloka ring complex revisited: palaeomagnetism and rockmagnetism

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SUMMARY

The isotopically dated Lower to Middle Devonian Sabaloka complex (16.3°N, 32.6°E) in Northern Sudan, for which a palaeomagnetic pole position close to the present geomagnetic pole had been derived in a previous investigation, has been revisited. The formerly determined direction of remanence is shown to be of secondary origin (possibly due to recent or Pleistocene weathering, as shown by ore microscopy and a negative conglomerate test) and a further component of magnetization is detected at several sites. Rockmagnetic studies give strong arguments in favour of a primary origin of the newly found remanence direction. In this paper the palaeomagnetic results (site means) of the two laboratories (Munich and Leeds) for the Sabaloka complex have been combined yielding $D = 340.1°$, $I = -43.9°$, $N = 15$, $k = 20.6$, $\alpha_{95} = 8.0°$. This gives a palaeomagnetic South Pole position (site mean) at 42.0°S and 55.7°E which is discussed in the context of other Palaeozoic pole positions of Gondwana.

Key words: African apparent polar wander path, palaeomagnetism, rock magnetism, Sabaloka, Sudan.

INTRODUCTION

The Palaeozoic portion of the apparent polar wander path of Africa and of Gondwana as a whole is still ill-defined. This is due to the general sparsity of data, their often poor age control and unclear tectonic positions of the investigated rock sequences. Especially striking up to now was the apparent contradiction between the Late Precambrian age determined on the basis of the palaeomagnetic pole position for the Sabaloka complex in Northern Sudan, as determined in first pilot studies by Briden (1973) and Cavanagh (1979), and considerably younger isotopic ages. Therefore a new detailed palaeomagnetic study was planned and carried out to resolve this controversy. A first and brief account of the new palaeomagnetic results for the Sabaloka complex are given in Saradeth (1986) and Saradeth *et al.* (1989). In the meantime the sample collections acquired by Briden and Cavanagh in the early 1970s were re-examined carefully and the new results were confirmed independently on their sample collection as well. The complex structure of the remanence, consisting of several components of different intensities (sometimes just above the detection level of modern instrumentation), as well as the complicated geologic and tectonic position made it desirable to combine the work of the two groups to provide a better basis for the palaeopole determination of the Sabaloka complex. In this paper the combined results of the two laboratories in Munich and Leeds will be presented.

GEOLOGY

The Sabaloka complex (16.3°N, 32.6°E) is one of the best exposed and most easily accessible ring complexes in the Sudan, situated astride the river Nile some 80 km north of Khartoum. It was subject to a number of studies: Delany (1955, 1958) gave a first geologic description of the ring structure and a detailed account of the geology, structure and petrography is presented by Almond (1977). The Sabaloka complex is a cauldron of elliptical shape with dimensions 15 by 20 km. The whole complex is bordered by a ringdyke as shown in the geological sketch map (Fig. 1). The most prominent topographic feature is an approximately round plateau with a diameter of about 11 km, that stands some 200 m up against the surrounding plain of gneisses and older granites. The River Nile has cut through this plateau and at the end of the canyon, the sixth cataract, called the Sabaloka Cataract, is situated (Fig. 1). Nubian sandstones cover parts of the complex in the northwest as well as Jebel Rauwiyan, a hill immediately to the southwest of the plateau. The plateau consists of acidic lavas and ignimbrites. To the south and west the plateau is

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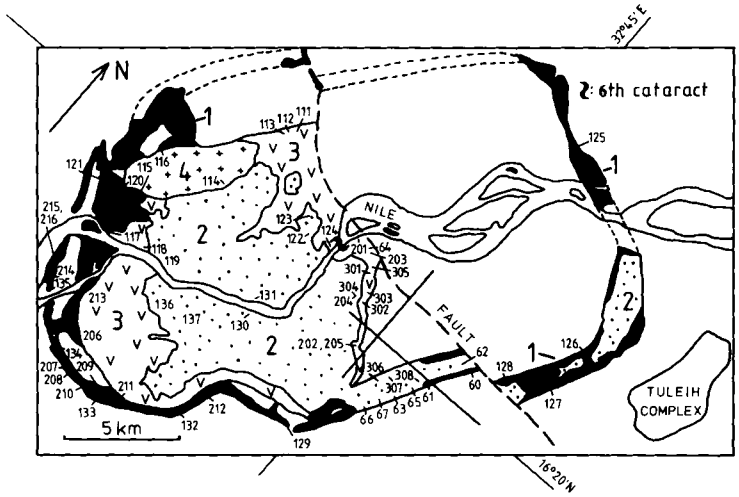


Figure 1. Geological sketch map of Sabaloka ring complex near the sixth cataract outlining the major lithological units (modified after Harris *et al.* 1983). (1) Ring dyke. (2) Ignimbrite. (3) Lavas, agglomerates. (4) Granite. Sampling sites in the Sabaloka ring complex are indicated as small numbers which coincide with the site numbers of Table 2. *Sites shown in Fig. 1 but not listed in Table 2 yielded no results.*

surrounded by hills of the granitic ringdyke. The fracture zone of the ringdyke disappears in the northwest under the sediment cover of the Nubian series. In the northeast, however, for some distance from the plateau, the continuation is indicated by low lying bouldery hills of granite. East of the plateau, the fracture zone consists of a sequence of volcanic rocks which have been preserved by subsidence in a graben structure. A large E–W trending post-Nubian fault dissects the cauldron and causes some offset of the whole structure. As the geologic sequence shows (Table 1a) an early volcanic cycle can be distinguished from a later intrusive cycle.

The subsidence of the central caldera consisting of basement and up to 2 km thick volcanics took place along the outer zone of ring fractures. Two main episodes of subsidence resulted in the formation of the caldera. At present the central part has risen through relief inversion. The bedding of the rhyolite lavas can be determined by flow featuring and flow banding, but there is no possibility to differentiate the original but not necessarily horizontal position from later tilting. The same holds true (Almond 1977) for the ignimbrite, where the tectonic position can be determined by eutaxitic foliation and columnar jointing. The volcanics are now dipping with inclinations up to 30° towards the centre of the caldera. The contact of the ringdyke with the surrounding host rock is very steep (70°–80°).

There have been several isotopic studies on this complex

Table 1. (a) Geologic succession of Sabaloka complex (simplified after Almond 1977). (b) Compilation of available isotopic age determinations for the Sabaloka complex; wr: whole rock data.

(*a*)

Nile silts, alluvial fans, aeolian sands		Cainozoic
----------	unconformity	
Nubian Series		Cretaceous
----------	unconformity	
ringdyke: porphyritic microgranite		Paleozoic
ignimbrite, micagranite		
upper rhyolite lavas		
Cataract Formation of volcanoclastic rocks		
lower rhyolite lavas		
basic lavas		
----------	disconformity	
subvolcanic sediments		Lower Paleozoic
----------	unconformity	
basement		Archaen or Proterozoic

(*b*)

Method	Age (Ma)	Lithology	Reference	Remarks
K/Ar	540±2	microgranite	Almond (1967)	wr
K/Ar	476±25	microgranite	Almond (1967)	wr
K/Ar	337	microgranite	Vail & Rex (1971)	wr
Rb/Sr	352±10	upper rhyolite	Cavanagh (1979)	wr, 6-point isochron
	339±20	same data, omitting one outlying point		wr, 5-point isochron
Rb/Sr	357±19	microgranite	Cavanagh (1979)	wr, 7-point isochron
Rb/Sr	399±7	rhyolite, porphyry	Barth et al. (1983)	wr, 3-point isochron
K/Ar	370	rhyolite	Kreuzer (1983) cited in Barth et al. (1983)	wr, preliminary result
Rb/Sr	383±14	micagranite	Harris et al. (1983)	wr, 4-point isochron
K/Ar	395±10	metaquartzite	Saradeth et al. (1989)	wr, etched

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(Table 1b). Whole rock K/Ar ages have been reported from the microgranite ranging from 337 to 540 Ma (Almond 1967; Vail & Rex 1971), and from contact metasediments (quartzites) at 338–402 Ma (arithmetic mean value: 395 ± 10 Ma; Saradeth *et al.* 1989). Cavanagh (1979) published whole rock Rb/Sr isochrons from the microgranite at 357 ± 19 Ma, and from the rhyolite at 352 ± 10 Ma, although Harris, Duyverman & Almond (1983) point out that the latter is biased by a single outlying point and that 339 ± 20 Ma may be a better estimate. Harris *et al.* (1983) also report their own Rb/Sr isochron on the mica granite at 383 ± 14 Ma, while Barth *et al.* (1983) obtained a Rb/Sr age of 399 ± 7 Ma. The weight of isotope age evidence thus points to a Lower to Middle Devonian age for the ring complex.

SAMPLING

For this study the sample collections from the palaeomagnetic working groups in Munich, which were acquired in two field trips during spring 1983 (sites 60–67) and fall 1985 (sites 111–137) were combined with the samples collected by the Leeds group in 1970 (sites 301–308) and in 1974 (sites 201–216). Details of the sampling sites (lithologies, tectonic correction data and so on) are given in Saradeth *et al.* (1989), Briden (1973) and Cavanagh (1979) and will not be repeated here. In total, 58 sites were investigated. Sampling focused on the ringdyke granite, which is regarded as the latest phase of activity and is not affected by subsidence, and on the ignimbrite, where structural control is fairly easily achieved. In addition, samples from the rhyolites, granites, an agglomerate, a felsic dyke and contact metamorphosed sediments are also examined. All sampling sites irrespective of their palaeomagnetic result are indicated in the geological sketch map (Fig. 1).

PALAEOMAGNETIC MEASUREMENTS

The first measurements in Leeds (Briden 1973; Cavanagh 1979) were carried out using a PAR spinner magnetometer. The NRM of these samples could not be completely destroyed by AF demagnetization. This collection and some other hitherto untreated samples were available for the later thermal demagnetization experiments.

The new results from both groups were obtained using fluxgate spinner magnetometers (Munich, Leeds) and a cryogenic magnetometer (Leeds) and stepwise thermal demagnetization was done in shielded furnaces. The characteristic remanence directions CARM were determined using Zijderveld diagrams of individual specimens (least-square fits) and the analysis of vector differences between steps of progressive thermal and AF demagnetization, respectively.

PALAEOMAGNETIC RESULTS

Details of the palaeomagnetic results (CARM) are listed in the Table 2(a) and 2(b). The NRM data can be taken from Briden (1973), Cavanagh (1979), Saradeth (1986), Saradeth & Soffel (1989) and Saradeth *et al.* (1989) and will not be repeated here because no conclusions were drawn from them. In total, the following four components of magnetization in both sample collections can be discriminated besides a short term acquired VRM component.

(a) A northerly downwards (N+) component as shown in Fig. 2(a). This component is carried by a highly coercive mineral and shows blocking temperatures up to 650 °C.

(b) A southerly upwards (S−) component antiparallel to the N+ component. This component is either the only predominant component or is superimposed by a less stable N+ component. This component is also characterized by high blocking temperatures up to 650 °C (Fig. 2b). Optimum cleaning of 40 sites yields the N+/S− direction of magnetization with a mean of $D = 359.5°$, $I = +29.3°$, $N = 40$, $R = 37.7$, $k = 16.6$, $\alpha_{95} = 5.4°$ and a palaeopole position of 250.2°E, 89.2°N, $dp = 3.3°$, $dm = 6.0°$. This is, within the limits of error, the same direction as detected by Briden (1973) and Cavanagh (1979). The mean pole is only 1° from the present pole of rotation i.e. this component of NRM corresponds to the local direction of the geocentric axial dipole field.

(c) In addition, a northerly upwards (N−) component is documented at some sites (see Table 2b). This component is in most cases of very small intensity and masked by the other components of magnetization. This remanence is characterized by higher stability versus cleaning and goes easily undetected due to the strong secondary overprint (Fig. 2c).

(d) There are also hints for the existence of a southerly downwards component (S+), antiparallel to the above cited N− component. In most cases, where there were signs of a S+ component, a stable endpoint during cleaning was not achieved due to the very low signal-to-noise ratio, and only at one site (211) was the S+ component clearly separated from others (Fig. 2d) in many specimens.

For these reasons, only one site with statistically significant reversed S+ directions (site 211) has been combined with the N− directions in computing the mean direction (Table 2b), while the S+ directions of specimens from sites 130, 201, 203, 206, 210 and 216 were not used because they lacked of the minimum statistical requirements. The results of 15 sites (14 normal, one reversed) yield a mean for this component of $D = 340.1°$, $I = -43.9°$, $N = 15$, $R = 14.3$, $k = 20.6$, $\alpha_{95} = 8.0°$ with a corresponding pole position (mean of all site mean pole positions) at 42.0°S, 55.7°E, $A_{95} = 7.3°$. Fig. 3 shows this pole position together with other Gonwana poles (see Table 3) of the time interval Ordovician to Carboniferous.

ROCKMAGNETIC MEASUREMENTS

The stability and origin of the remanence was determined by thermomagnetic measurements, IRM acquisition curves and ore microscopy. Examples of these measurements are shown in Fig. 4. The thermomagnetic curve (J_s/T) of a powdered specimen from site 62 (Fig. 4a) yields evidence for magnetite and maghemite. Their presence is indicated by a Curie temperature of 560 °C and a sharp drop of J_s between 300 °C and 400 °C in combination with a 'kink' at around 150 °C, respectively. Haematite could not be identified by these measurements. However, there is evidence for haematite due to high coercive forces as shown during IRM

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Table 2. Results of the combined sample collection from Munich (sites 60–77, 111–137) and Leeds (201–216, 301–308) for Sabaloka complex (16.3°N, 32.6°E), omitting all sites with scattered and inconsistent directions. The following data are presented: site number; stability range of alternating field (mT) and thermal demagnetization (°C), respectively; n: number of samples; N: number of cores; R: resultant vector; D: declination; I: inclination; k, K: precision parameters; α_{95}, A_{95}: semi opening angle of the cone of confidence; coordinates of the virtual geomagnetic southpole (VGP) for each site in °S and °E, respectively; dp, dm: semi-minor and semi-major axis of the polar error ellipse at the 95 per cent probability level.

2a: Data of the secondary remanent magnetization.

Site No.	Range (°C,mT)	n/N	Secondary remanence component D	I	k	α_{95}	Remarks
60	< 560°/100mT	12/8	350.9	14.8	46.0	6.5	
61	< 250°	10/8	349.1	-25.8	15.0	12.9	
62	< 350°/100mT	17/9	356.8	11.3	51.2	5.1	
63	< 500°	16/9	352.3	25.8	102.1	3.7	
64	100mT	11/8	5.6	28.6	60.2	5.9	
65	< 650°/100mT	12/9	347.7	25.3	160.7	3.4	
66	< 250°/100mT	12/8	352.1	5.2	60.5	5.7	
67	< 660°/100mT	17/9	186.2	-30.3	86.2	3.9	16 reversed
112	< 560°	8/7	357.6	25.8	47.3	8.1	
116	< 640°	6/5	345.0	44.6	5.2	32.5	5 reversed
117	< 550°	5/5	12.0	13.7	2.5	--	1 reversed
118	< 510°	9/8	353.5	54.2	18.7	12.2	
120	< 610°	7/6	5.1	48.9	12.6	17.7	
121	< 620°	18/8	356.6	-6.5	15.1	9.2	
123	< 620°	6/5	336.1	37.3	11.3	20.8	
124	< 640°	13/10	349.0	31.5	10.8	13.2	
125	< 650°	6/5	24.9	31.9	11.3	20.8	
126	< 670°	4/4	3.1	33.1	6.1	40.5	
127	< 650°	7/6	9.7	55.2	4.3	32.8	2 reversed
128	< 650°	7/7	349.3	14.4	21.9	13.2	
129	< 610°	8/6	352.8	35.5	15.6	14.5	4 reversed
132	< 550°	7/6	8.4	27.1	23.1	12.8	3 reversed
133	< 550°	5/5	1.4	26.0	11.1	24.1	1 reversed
134	< 650°	9/5	5.5	40.8	7.0	20.9	
135	< 650°	7/6	351.5	7.5	8.4	22.0	
136	< 650°	9/7	359.5	28.3	16.4	13.1	2 reversed
203	< 111°	1/1	9.5	47.8	-	-	
206	up to 670°	4/2	352.9	26.2	43.7	-	
207	up to 670°	1/1	220.8	-23.9	-	-	reversed
208	up to 670°	4/3	7.7	32.3	71.8	10.9	
210	up to 670°	2/2	185.0	-30.3	31.5	-	reversed
211	< 335°	6/5	355.1	42.7	12.5	19.7	
212	up to 670°	10/6	9.8	42.2	17.5	11.9	
216	165-560°	2/2	350.9	43.4	40.1	-	
301	< 214°	1/1	4.4	23.2	-	-	
302	< 590-650°	13/7	24.4	55.9	10.2	13.6	
304	< 250°	7/6	13.5	54.2	9.6	20.6	2 reversed
305	106-400°	1/1	353.2	20.8	-	-	
306	< 365°	3/2	350.1	8.0	26.0	-	
307	< 210°	1/1	342.7	27.1	-	-	
Mean remanence direction of all sites:							
R=37.66		40	359.5	29.3	16.6	5.4	
Mean pole position: 89.2°N, 250.2°E, dp=3.3°, dm=6.0°							

acquisition [specimen from site 64, Fig. 4(b)] and AF treatment in combination with blocking temperatures up to 650 °C (see Fig. 2). An IRM acquisition curve for a specimen with mainly magnetite and/or maghemite from site 119 is shown in Fig. 4(c).

Ore microscopy gives a strong argument in favour of the presence of primary haematite. Grains with a haematite rim and an inner core of pseudobrookite, pseudomorphous after titanomagnetite, indicate an origin through oxidation at high temperatures, conditions prevailing only a short time after extrusion [specimen from site 66, Fig. 4(d)]. A secondary origin of this haematite by weathering or other processes can be excluded according to El Goresy (1986, personal communication). Haematite of that sort is abundant in those specimens where a N−/S+ component (Table 2b) could be isolated. But at many sites the entire ore content is strictly of secondary origin due to recent or Pleistocene weathering as result of martitization or maghemization. Fig. 4(e) shows a strongly altered former titanomagnetite grain (specimen from site 63) consisting mostly of maghemite (medium grey) and secondary haematite (light grey). These two ore phases seem to carry the secondary N+/S− remanence component which agrees rather well with the present or subrecent field direction in the sampling area (Table 2a).

DISCUSSION OF THE PALAEOMAGNETIC RESULT AND CONCLUSIONS

The N+/S− component of remanence (Table 2) is not statistically different from the present geocentric dipole field. It can be assumed to have originated recently or during the Upper Tertiary (Pleistocene) by chemical weathering and is carried by both haematite and maghemite.

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Table 2. (*continued*)

2b: Data of the primary remanence CARM, core corrected. The mean VGP coordinates were calculated from the individual VGP data, which were computed from core corrected remanence directions. *: sites taken for the computation of the mean CARM direction.

Site No.	Range (°C,mT)	Characteristic remanence CARM core corrected n/N	D	I	k	α95	Remarks	VGP (°S)	(°E)
61*	250-600°	8/8	4.8	-68.6	14.5	15.1		21.7	29.4
62*	400-600°	15/9	339.6	-57.5	27.4	7.4		32.3	51.5
66*	250-560°	11/7	351.3	-33.7	13.7	12.8		54.2	46.8
113*	25-60mT	11/7	354.5	-36.0	22.9	9.7		53.3	41.3
119*	300-560°	13/8	354.5	-57.8	237.8	2.7		35.0	37.9
122*	270-640°	17/8	344.3	-32.6	131.6	3.1		52.6	57.7
130	scattered, indication for a reversed direction								
131*	450-640°	9/7	326.7	-11.3	15.9	13.3		50.4	91.6
134*	50-80mT	5/3	327.3	-51.6	7.5	30.0		32.2	65.3
137*	420-600°	13/8	319.7	-19.1	40.6	6.6		42.3	92.1
201	scattered, indication for a reversed direction								
203	scattered, indication for a reversed direction								
206	› 640°	1/1	126.4	+48.6	-	-	reversed	-	-
210	› 310°	1/1	125.5	+29.7	-	-	reversed	-	-
211*	› 300°	5/4	169.8	+40.3	11.1	24.0	reversed	49.5	47.1
216	‹ 636°	2/2	167.6	+28.9	22.6	-	reversed	-	-
301*	310-650°	2/2	327.6	-57.1	85.8	-		28.0	61.4
304*	380-650°	7/5	352.5	-37.4	25.7	12.1		44.1	52.1
306*	188-355°	1/1	327.8	-48.1	-	-		34.9	67.2
307*	300-680°	2/2	346.4	-47.4	31.6	-		43.3	49.1
308*	310-660°	4/4	341.0	-52.6	119.3	8.5		37.3	52.6
Means:		15 R=14.32	340.1	-43.9	20.6	8.0		42.0 A_{95}=7.3° N=15,R=14.43 K=23.54	55.7

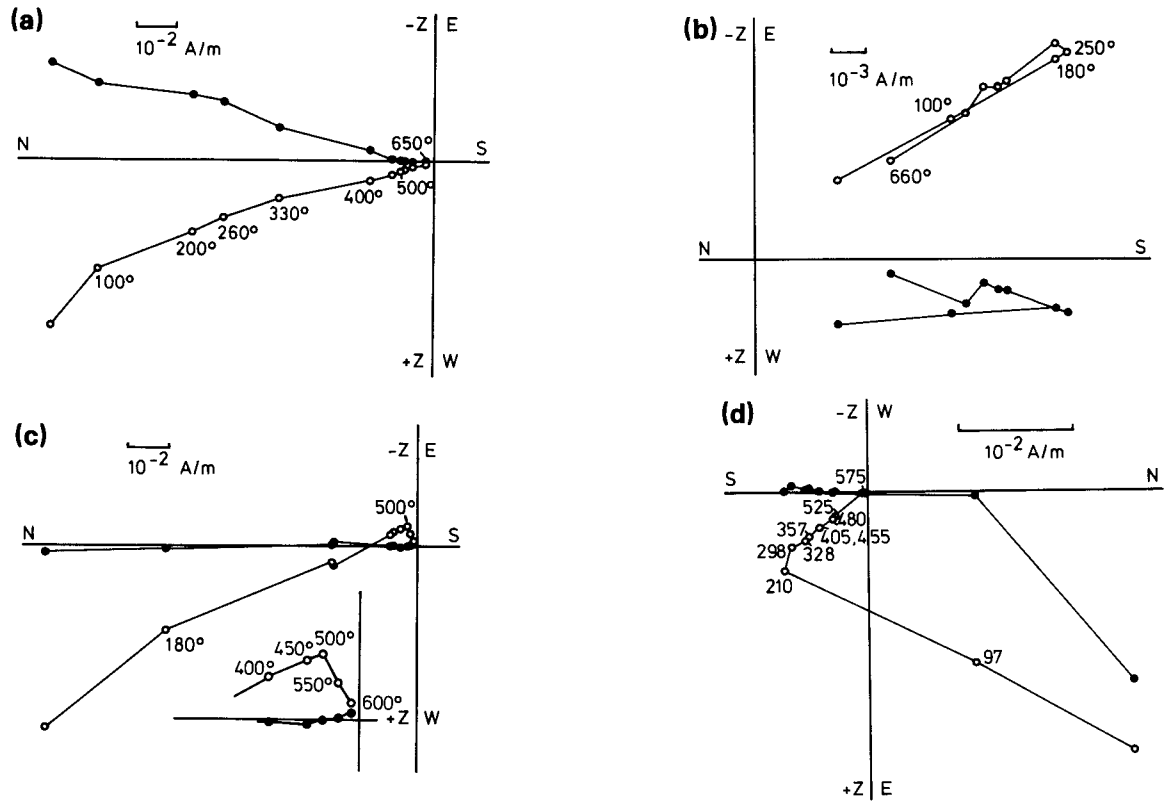


Figure 2. Vector diagrams showing the demagnetization behaviour of samples during thermal demagnetization. Examples for AF demagnetization can also be taken for instance from Saradeth (1986). Closed symbols: projection into horizontal plane, open symbols: projection into vertical N–S plane. (a) Sample from site 129. The presence of a northerly downwards (N+) component of remanence is revealed during thermal treatment. (b) Sample from site 67. Thermal demagnetization up to 250 °C completely destroys the northerly downwards (N+) component of magnetization, then an antiparallel southerly upwards (S−) component is revealed. (c) Thermal demagnetization of a sample from site 62. Insert: Close-up of the high-temperature range of thermal demagnetization. Treatment with temperatures above 500 °C clearly indicates a well-defined northerly upwards (N−) component of remanence. (d) Thermal demagnetization of a sample from site 211 indicating the possible presence of a southerly downwards (S+) component.

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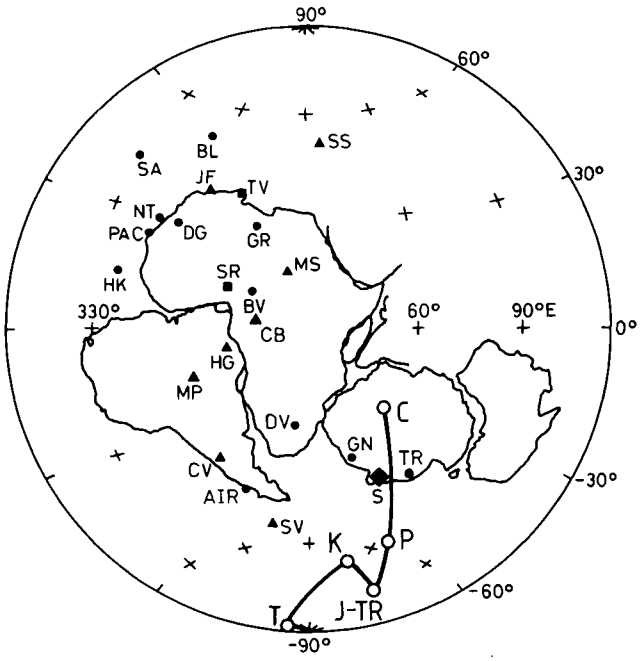


Figure 3. The Devonian Sabaloka pole position S (large diamond) in context with other Gondwana poles of the time interval Ordovician to Carboniferous. For abbreviations of poles see Table 3. Dots: data from Africa. Triangles: data from Australia. Squares: data from Eastern Antarctica. The rather simplified APWP for Africa (south poles) from present to Carboniferous (based on a compilation by Piper 1987) is shown as a bold line using conventional abbreviations for the geologic formations. The course of the APWP during the Middle and Lower Palaeozoic is still a matter of dispute and the various published paths (see text) are not shown.

This is confirmed by a negative conglomerate test. At site 124, the coarse volcanoclastic rocks of the Cataract Formation were sampled. The mostly crudely bedded volcanic breccias, in which a few blocks reach 2 m in diameter, consist almost exclusively of rhyolite lava. Almond (1977) considers them to be deposits from sheet floods or water-rich mudflows. Ten rhyolite blocks were sampled. AF treatment yields the northerly downward direction quite well-defined with a precision parameter of $k = 10.8$. This means that the conglomerate test is negative; the remanence vectors are not randomly distributed as should be expected for a primary origin of magnetization. Additional thermal treatment did not yield any more well-defined further components of magnetization. The existence of two almost antiparallel components N+ and S− (Table 2a) also indicates the slow acquisition of the secondary magnetization during at least one field reversal.

The N− component (Table 2b) was clearly found at 14 sites and is mainly carried by primary haematite, as identified in ore microscopy. Haematite originates in volcanics through oxidation during cooling. Zijderveld *et al.* (1970) also describe primary haematite in ignimbrites as main carrier of stable remanence in the Bolzano quartz porphyries. It was only possible to detect the existence of the N− component at sites where it was not completely or to a large extent replaced by the secondary overprint. This component is regarded as primary. At site 211, and presumably at several other sites (130, 201, 203, 206, 210, 216), there are indications of a reversed component S+, antiparallel with respect to N−, of which only the statistically most reliable direction of site 211 was used for the computation of a mean. The presence of the S+ component in addition to the N− component shows that the acquisition of this CARM extended over a period with at least one polarity reversal.

It now appears that the northerly upward directions found by Briden (1973) at sites 307 and 308 were primary magnetizations. The 'positive fold test' of Briden (1973) in which these directions were corrected by restoring tilted lavas within the ring dyke to the horizontal, now seems to have been fortuitous. In retrospect, it was not a convincing test, as the requirement of Fisher distribution of directions *in situ* and corrected for tilt was not fulfilled, and the success of the test hinged on a single section of trachytes with anomalous dip.

The mid Palaeozoic palaeomagnetic poles for Africa and Gondwana (Table 3) are not so well documented as one might expect from the published data of polar wander paths of Africa in textbooks and review papers. The pole position for Africa obtained by Hailwood (1974) from the Msissi Norite in Morocco was until recently regarded as the most reliable Devonian pole. However, it has been shown recently by Salmon *et al.* (1986) that the Msissi Norite is in fact of Upper Jurassic to Lower Cretaceous age and that its magnetization is far more complex than previously assumed. In addition, most palaeomagnetic poles for the Middle Palaeozoic of Gondwana stem from the Tasmanian terrane in SE Australia. The palaeogeographic relation of the Lachlan fold belt with respect to cratonic Australia is also unclear during the Middle Palaeozoic (Crook 1980), and this holds also for other fragments of Gondwana. Because of this difficulty, several apparent polar wander paths have been suggested for Gondwana for the Palaeozoic by McElhinny & Embleton (1974), Schmidt & Morris (1977), Morel & Irving (1978), Goleby (1980) and Livermore, Smith & Briden (1985), none of which can be favoured on the basis of the new result from the Sabaloka complex. In order to avoid confusion we preferred to draw in Fig. 3 no polar wander path between the more or less undisputed poles off SE Africa for the Carboniferous and off NW Africa for the Ordovician.

The interpretation of the Sabaloka pole position depends critically on the age of the N−/S+ direction, which is regarded as the CARM. The close agreement of the Sabaloka pole with the Middle Devonian Gneigura pole position (Kent, Dia & Sougy 1984) may be an indication for the CARM age to be equal or close to the lithologic age (Lower to Middle Devonian, Table 1b). However, this assumption would lead to phases of rapid drift between the Ordovician (with a pole position off NW Africa) and the Devonian (with a pole position off SE Africa) and a start of the Permian and Carboniferous quasistationary period already in the Devonian. In terms of palaeogeography, the northern border of Gondwana would have been separated from Laurasia in the Devonian only by a small ocean. This is supported by biogeography, favouring a small separation rather than a wide dividing ocean between the two landmasses (Scotese, Van der Voo & Barrett 1985). However, a Devonian CARM and a corresponding pole

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Table 3. Selected Ordovician to Devonian palaeopole positions from Africa, Australia and Antarctica in a Gondwana Reassembly.

Formation	Symbol	age	Lat. [°N]	Long. [°E]	k	a_{95} dp,dm	Reference
Africa							
Damara granites	DG	458 ± 8[2]	27	351	-	5	Corner & Henthorn (1978)
Hook intrusives	HK	517 ±18[2]	14	336	13	36	Brock (1967)
Ntonya Ring Complex	NT	485 ± [3]	28	345		1,2	Briden (1968)
Graafwater Formation	GR	O1[4]	28	14		7,12	Bachtadse et al. (1987)
Blaubeker Formation	BL	471 ±26[2]	51	352	-	11,20	Kröner et al. (1980)
Salala Ring Complex component B	SA	~460[2]	40	330	-	8,12	Bachtadse & Briden (1989)
Pakhuis/Cedarberg Formations	PAC	Ou/S1[4]	25	343	-	12,21	Bachtadse et al. (1987)
Air intrusives	AIR	435[3]	-43	9	50	6	Hargraves et al. (1987)
Bokkeveld Group	BV	Dm[4]	10	15	-	9,11	Bachtadse et al. (1987)
Gneiguira Formation	GN	Dm[4]	-35	44	-	3,6	Kent et al. (1984)
Dwyka Varves	DV	Cu[4]	-26	26	52	10	McElhinny & Opdyke (1968)
Tanzanian redbeds	TR	P-C[4]	-40	64	11	16	McElhinny et al. (1968)
Sabaloka complex	S	Dm[2,3]	-42	56	24	7	this study
Australia							
Jinduckin Formation	JF	O1[4]	36	359	10	11	Luck (1972)
Stairway Sandstone	SS	Om[4]	51	34	25	9	Embleton (1972)
Canning Basin	CB	Du[4]	2	16	62	8	Hurley & Van der Voo (1987)
Hervey Group[1]	HG	Du[4]	-5	8	-	8,16	Li et al. (1988)
Mugga Mugga porphyry[1]	MP	414±8[2]	-16	2	-	6,9	Briden (1966); revised by Goleby (1980)
Snowy river volcanics[1]	SV	D1[4]	-54	15	-	11,14	Schmidt et al. (1986)
Comerong volcanics[1]	CV	Dm[4]	-34	2	42	7	Schmidt et al. (1987)
Eastern Antarctica							
Sor Rondane intrusives	SR	~469[2]	11	8	-	5,6	Zijderveld (1968)
Taylor Valley dykes	TV	460±7[2]	36	9	-	6,11	Manzoni & Nanni (1977)

Palaeomagnetic poles from Australia and Antarctica have been transferred into African coordinates using the rotation parameters given by Lawver & Scotese (1987). Data have been selected using the minimum criteria introduced by McElhinny (1973). Southamerican paleozoic poles have not been included because of uncertain rock and magnetization ages. Whenever necessary isotopic ages have been recalculated using the appropriate decay constants (Steiger & Jäger 1977). a_{95}: semi angle of the cone of confidence about the palaeopole position at the 95% probability level (Fisher 1953). Lat,Long: latitude,longitude of the palaeomagnetic pole; dp,dm: semi axis of the oval of 95% confidence about the mean pole; k is the Fisherian precision parameter associated with the means. O: Ordovician; S: Silurian; D: Devonian; P-C: Permo-Carboniferous; l, m, u: lower, middle, upper.

1) palaeopole positions from the (allochthonous?) eastern Australian Lachlan Fold Belt.

2) Rb/Sr age

3) K-Ar age

4) stratigraphic age

position would also mean that the poles situated in central Africa, referred to Middle to Upper Devonian (Fig. 3) would have to be rejected.

Alternatively there are reasons in favour of a CARM age considerably younger than the mean lithologic age which itself should be used with caution. Complex cauldrons like the Sabaloka complex with several cycles of volcanic and intrusive activity (Table 1a) and with several cycles of subsidence can be expected to show magmatic and hydrothermal activity over a considerable time span. This and the different blocking temperatures associated with the age dating methods used (K/Ar and Rb/Sr, respectively) is reflected by the range of isotopic ages. The palaeomagnetic sampling was focused on the latest phase of the development of the complex. Samples from the ring dyke and the ignimbrites were favoured because of their palaeomagnetic suitability as well as in order to avoid unwanted later tilting and internal tectonic movements of

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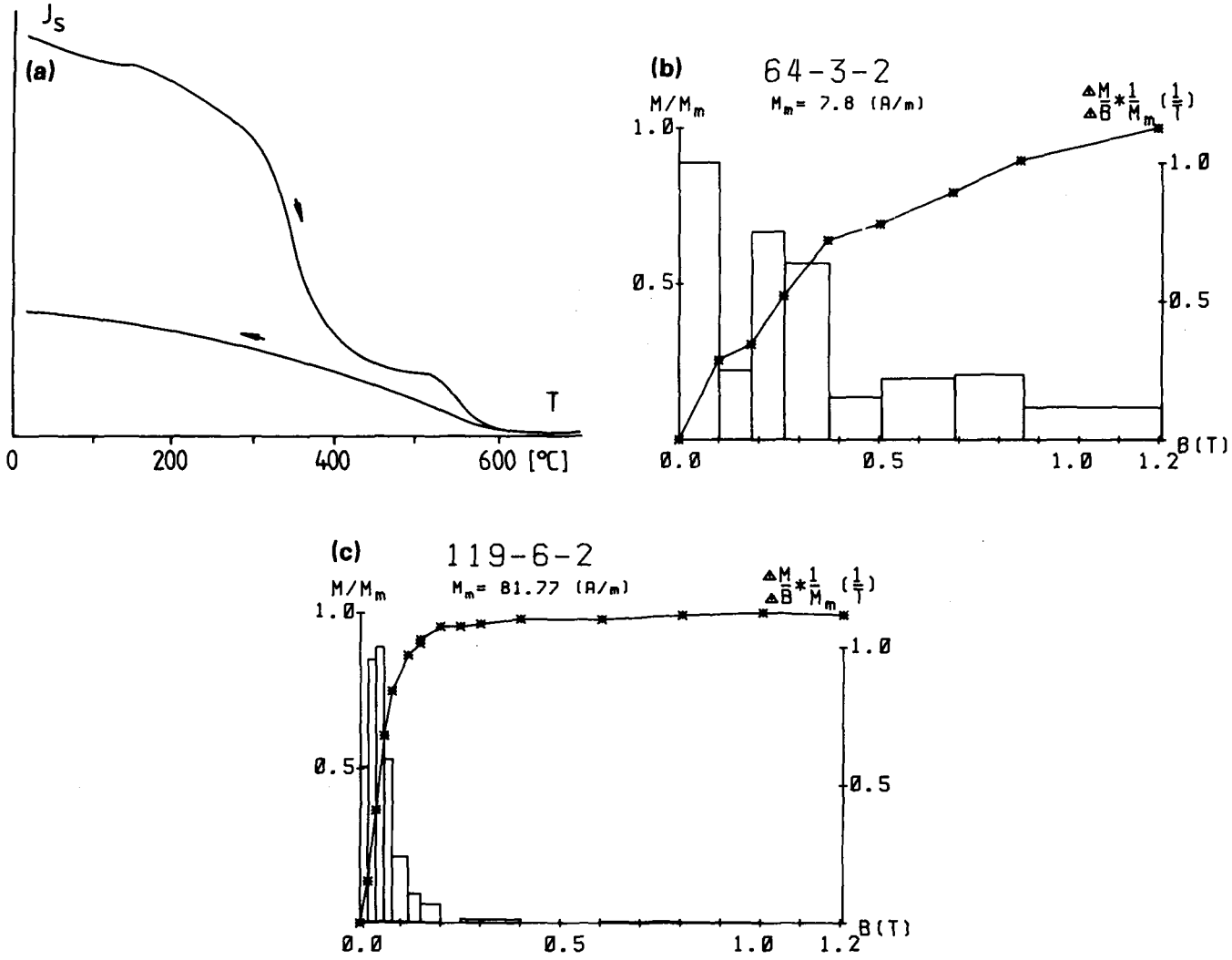


Figure 4. Rockmagnetic measurements and ore microscopy. (a) Thermomagnetic curve (J_s/T) measured in a field of 0.5 T of a powdered specimen from site 62. The Curie temperature at about 560 °C is indicative for a ferrimagnetic phase close to magnetite, while the drop of the curve near 350 °C and the 'kink', near 150 °C point to maghemite. Haematite could not be detected in the J_s/T curves. (b) IRM acquisition curve and coercivity spectrum for a specimen from site 64. No saturation was obtained for fields up to 1200 mT showing the presence of haematite. (c) IRM acquisition curve and coercivity spectrum for a specimen from site 119. Saturation was obtained with a field of 300 mT indicative for magnetite and/or maghemite as the dominant ferrimagnetic ore phase. (d) Polished section of a specimen from site 66 showing an ore grain pseudomorphous after a cubic spinell phase (titanomagnetite) with a white haematite rim round an inner darker core consisting of pseudobrookite. The former titanomagnetite was highly oxidized at high temperatures (El Goresy 1986, personal communication). This haematite phase is believed to carry the N−/S+ CARM direction (see text and Table 2b). (e) Polished section of a specimen from site 63. The grain consists of a mixture of maghemite (medium grey) and haematite (light grey). Limonite can be seen at the rim of many ore grains as well as in the rock matrix. This former titanomagnetite has been strongly altered by low-temperature oxidation (El Goresy 1986, personal communication). Its oxidation products maghemite and haematite are most likely the carriers of the secondary N+/S− component (see text and Table 2a).

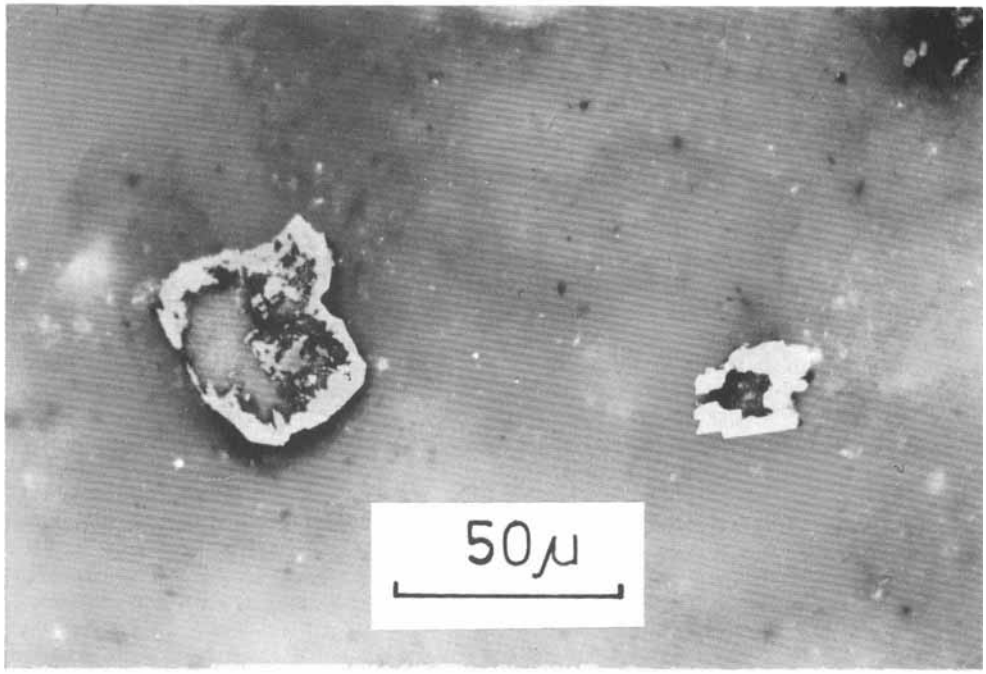
the complex. In contrast to that, many samples for isotopic age determinations were also taken from a wide range of lithologies among also older rocks of the complex. Because of this we regard the youngest of the K/Ar ages (337 Ma) and of the Rb/Sr ages (339 ± 20) as an upper limit for a more likely age of the N−/S+ CARM component. This Early to Middle Carboniferous Sabaloka pole position fits better with the established part of the APWP from Carboniferous to recent (Fig. 3). With this interpretation the new Sabaloka pole cannot contribute to a better defined APWP of Gondwana in the Lower to Middle Palaeozoic. It also suggests that the Gneigura pole (Kent *et al.* 1984) may be biased by remagnetization.

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(d)



(e)

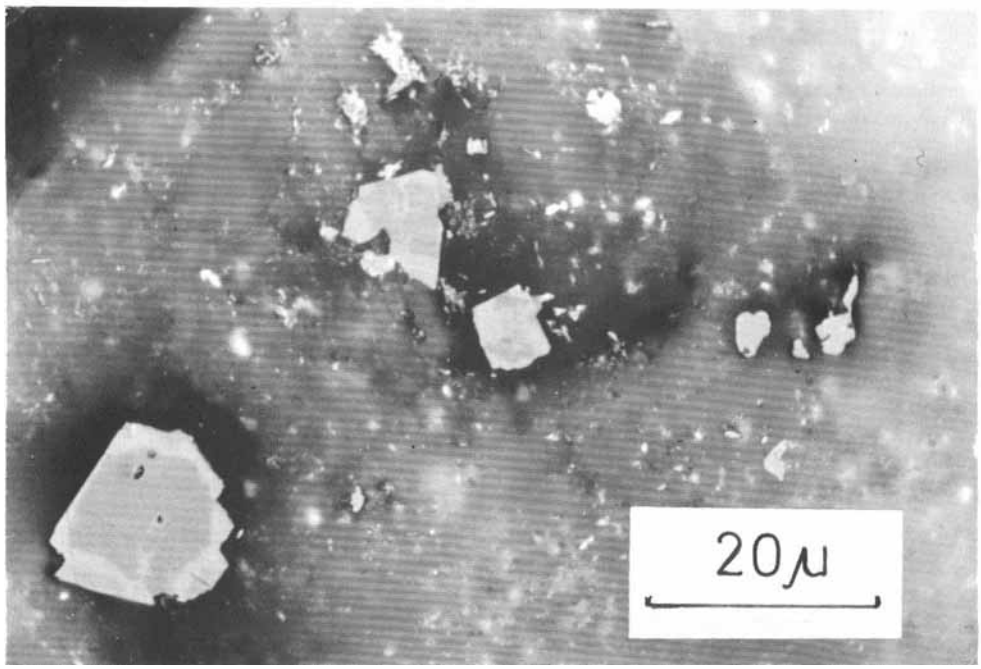


Figure 4. (*continued*)

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